Class Questions

Problem Statement

A discrete source generates an information sequence using the quaternary alphabet $\{a, b, c, d\}$ with probabilities:

$$\mathbb{P}(s = a) = \frac{1}{8},$$
$$\mathbb{P}(s = b) = \frac{1}{8},$$
$$\mathbb{P}(s = c) = \frac{1}{2},$$
$$\mathbb{P}(s = d) = \frac{1}{8}.$$

What is the entropy of the source?

Problem 2: Binary Phase-Shift Keying (BPSK)

Binary phase-shift keying (BPSK) is one of the simplest forms of digital modulation. Let $i[n]$ denote the sequence of input bits, $s[n]$ a sequence of symbols, and $x(t)$ the continuous-time modulated signal. Suppose that bit 0 produces A and bit 1 produces $-A$.

- Create a bit-to-symbol mapping table.
- BPSK is a type of pulse-amplitude modulation. Let $g_{\text{tx}}(t)$ denote the pulse shape and let

$$x(t) = \sum_{n=-\infty}^{\infty} s[n] g_{\text{tx}}(t - nT) \tag{2.16}$$

denote the modulated baseband signal. Let the pulse $g_{\text{tx}}(t) = \frac{1}{\sqrt{T}}$ for $t \in [0, T]$ and $g_{\text{tx}}(t) = 0$ otherwise.

Draw a graph of x(t) for the input bit sequence $i[0] = 1, i[1] = 1, i[2] = 0, i[3] = 1$ with $A = 3$ and $T = 2$, plot $x(t)$ for $t \in [0, 8]$.

Problem 3: Fourier Transform of a Triangular Pulse

Compute the Fourier transform of

$$\Lambda\left(\frac{t}{T}\right) = \begin{cases} 1 - \frac{|t|}{T} & |t| \leq T \\ 0 & \text{elsewhere} \end{cases} \tag{3.12}$$

using the fact that $\Lambda\left(\frac{t}{T}\right) = \frac{1}{T}\,\text{rect}\left(\frac{t}{T}\right) * \text{rect}\left(\frac{t}{T}\right)$.

Problem 4: Inverse Continuous-Time Fourier Transform (CTFT)

Compute the inverse CTFT of $x(2f)\cos(2\pi f t_0)$, where t_0 is constant.

Problem 5: Probability Density Function

Suppose that $f_x(x) = \frac{1}{2}$ for $x \in [0, 2)$ and is zero elsewhere. Calculate the mean and variance of x

Problem 6: Moving Average Random Process

Suppose that $s[n]$ is a real, IID WSS random process with zero mean and covariance $R_{ss}[k] = \sigma_s^2\delta[k]$. Suppose that $x[n] = s[n] + 0.5s[n-1]$, which is an example of what is called a moving average process. Compute the mean and covariance of $x[n]$, and determine if it is WSS.